Compact Laser MBE System

(Model: PAC-LMBE)

Operation Manual

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1 Introduction

1.1 About this document

This document, PLD operation manual, describes each procedure in accordance with deposition workflow. There are three kind of documents for the PLD system. One is the system manual that describes structure and function of each part/unit in the PLD system. Another is the operation manual. The last is maintenance manual.

Some units in the PLD system have occasions to injure operators or to damage the lab facility when user mistakes. See following chapters and other documents which show functions of individual units and parts in detail.

1.2 Caution

	Class 4 laser equipments are used in the PLD system. Keep away from the laser equipments and its beam path during laser emission. Wear protection glasses and clothes if user has to observe inside of the chamber or approach the laser beam path.
	X-ray is generated when an electron beam is irradiated to a substrate material with RHEED (Reflection High Energy Electron Diffraction) gun. Keep away from viewing ports which have no protection.
	Numerous electric power lines with large current or high voltage and sensor signal cables are wired at/to terminals or connectors. Do not touch the lines/cables while the PLD system runs. Shut down the electric power when user has to do cable works.
	As for an optional use of harmful or toxic gases, our original system does not equip any safety protections including a gas detector. On the occasion of use these kinds of gases, a leakage check and accompanying measure against accidents should be done thoroughly by the user.
	This system cannot sustain a normal operation under lack cooling water flow rate. An inspection of the compressed air and cooling water is regularly required.
	This system basically requires being electrified continuously. Connecting the system to a power line of insufficient capacity may cause an accidental electric power failure or system malfunction. The rating of electric power line should be sufficient.
	We do not undertake a maintenance service or a repairing in the condition that the equipment contains harmful, toxic substances or products and specified chemical substances etc. These substances should be completely removed prior to the services.

2 Check the PLD system before deposition

This chapter describes required operation at each deposition step. First it is assumed that following items are ready before deposition process.

(1) The PLD system with vacuum pumping lines including pumps and valves, the substrate heater/manipulator, the target manipulator, the process gas-line, and the ablation laser beam path for target material are properly assembled in accordance with description in the system manual and the factory setting.

(2) Electric connection of power supply cables and control signal lines are properly done.

(3) The PLD system runs ordinary status when user keeps non-stop (overnight) vacuum pumping operation. Or all alarm indicators/buzzers do not work after user's turning power on.

Keep ready for reviewing all documents for the PLD system, components, and the parts to resolve unknown or unexpected status during the deposition.

3 Workflow of the deposition in the PLD system

An epitome of deposition workflow is presented in table form on next page. User can use the table as a deposition check sheet. Notation of “4-x” at each column corresponds to following sections to describe detail procedure in use of PLD part.

PLD System: Deposition check sheet (Templete)

Date:		Operator:	
Purpose:			
Substrate:			
Targets (No.)			
Operation		Check	Remarks
1	Start up (Sec. 4-1).		(Main-ch pressure) 1.2 ×10- 6 Pa
2	Vent (L/L-ch) (Sec. 4-2).		
3	Setting substrate/target (Sec. 4-3).		
4	Vacuum pumping (L/L-ch) (Sec. 4-2).		
5	Transfer substrate/target (Sec. 4-4).		
6	RHEED setup/observation		
7	Process gas feeding (Sec. 4-5)		Pressure =
8	Substrate heating (Sec. 4-6).		Tsub =
9	Deposition with laser control (Sec.4-7).		Laser parameters:
10	Post-deposition treatment.		Stop sub. Heating / Stop process-gas flow.
11	Transfer substrate/target (Sec. 4-4).		
12	Vent (L/L-ch) (Sec. 4-2).		
13	Pick up substrate/target (Sec. 4-3).		
14	Vacuum pumping (L/L-ch) (Sec. 4-2).		
15	Shut down optional units. Vacuum pumping		Ablation laser, Laser diode PS, and Chiller. (Main-ch pressure) 1.2 ×10- 6 Pa

4 Procedure during PLD operation

4.1 Start the vacuum system up

Most of vacuum pumping procedure can be done by button click operation in the PC control software. And the pumps on/off status and the valves open/close status can be checked by illumination of the button indicator.

(Load-lock chamber)

(1) Check whether the gate valve GV1 is open or not.
When the valve GV1 is open, following pumping process makes high vacuum in both L/L chamber and main chamber.

(2) Close the access hatch.

(3) Turn on the pump RP3: run status of the RP3 is green in the button indicator.

(4) Open the valve FV3: open status of the FV3 is green in the button indicator.

(5) Turn on the pump TMP3: acceleration and normal rotation correspond green blink and green illumination of the button indicator. And check the display in the TMP3 controller.

(6) Close FV3 and open valve RV3 for roughing when vacuum pressure of the L/L chamber measured at TC3 exceeds set-points value. Or skip this process when two set-point indicators at TC3 light green color. Check open/close status in the button indicators of FV3 and RV3.

(7) Check pressure set-point indicators “SP1” and “SP2” at TC3 light in green. It is a condition to change vacuum pumping line in L/L chamber (about 10 Pa).

(8) Change vacuum pumping line. Close valve RV3 manually and open valve FV3. Check open/close status in the button indicators of FV3 and RV3.

(9) Open valve MV3: open status of the MV3 is green in the button indicator.

(10) Turn on a filament of IG3 vacuum gauge. Click “Fil 1/2” button of IG3: working status of the IG3 is green in the button indicator.

(Main chamber with RHEED differential pumping lines)

(11) Turn on the pump RP1: run status of the RP1 is green in the button indicator.

(12) Turn on the pump RP2: run status of the RP2 is green in the button indicator.

(13) Open the valve FV1 and valves FV20 and FV21: open status of the valves is green in the button indicator.

(14) Turn on the pump TMP1 and pumps TMP20 and TMP21: acceleration and normal rotation correspond green blink and green illumination of the button indicator. And check the display in the TMPs' controller.

(15) Open the valves MV11: open status of the MV11 is green in the button indicator.

(16) Open the valve MV12 manually.

(17) Open the valve MV10: open status of the MV10 is green in the button indicator.

(18) Open the valves MV20 and MV21 for RHEED e-gun differential pumping line: open status of the valves is green in the button indicator.

(19) Open the valve CV1 near capacitance manometer CDG10: open status of the CV1 is green in the button indicator.

(20) Turn on a filament of IG1 vacuum gauge. Click “Fil 1/2” button of IG1: working status of the IG1 is green in the button indicator.

(21) Close the valve GV if open.

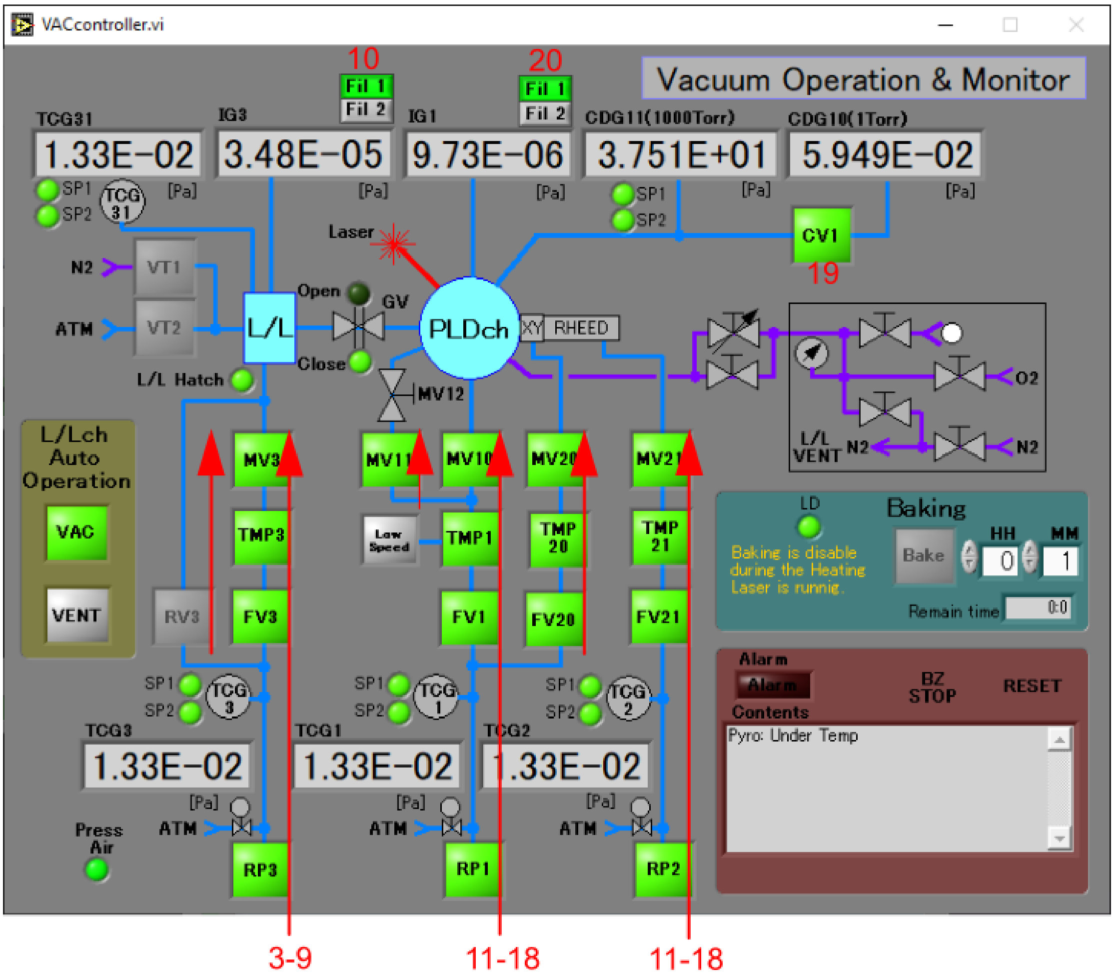

Figure 1:Vacuum pumping operation in PC control software.

4.2 Vent/vacuum pumping of load-lock chamber

Vent and vacuum pumping in the load-lock chamber can be automatically accomplished by clicking "VENT" or "VAC" button without each pump ON/OFF operation or valve open/close operation.

(Vent)

(1) Loose the screw at access hatch.

(2) Click "VENT" button: turning off a filament of IG3 vacuum gauge, closing MV3 and successive opening vent gas line are automatically done.

(3) Open access hatch.

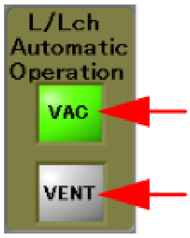

Figure 2:Automatic operation in L/L-ch.

(Vacuum pumping)

(4) Close access hatch.

(5) Click "VAC" button: vacuum pumping procedure with roughing and turning on a filament of IG3 vacuum gauge are automatically done.

4.3 Setting substrate holder/target holder

Substrate holders and target holders can be put on stage in the load-lock chamber as shown in Figure 3.

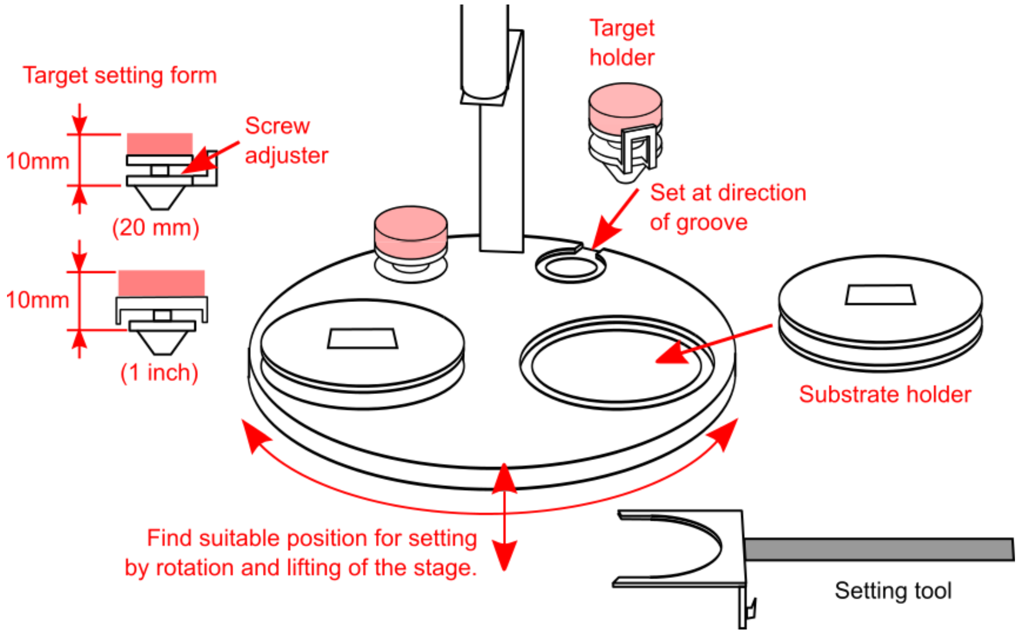

Figure 3:Substrate holder/taget holder setting in load-lock chamber.

4.4 Substrate holder/target holder transfer

(1) Catch a substrate holder/a target holder: handle the substrate holder or the target holder by two

kinds of setters, a folk for substrate holder or a hook for target holder at an end of the transfer-rod. Manipulation during catching the substrate holder or the target holder is shown in Figure 4.

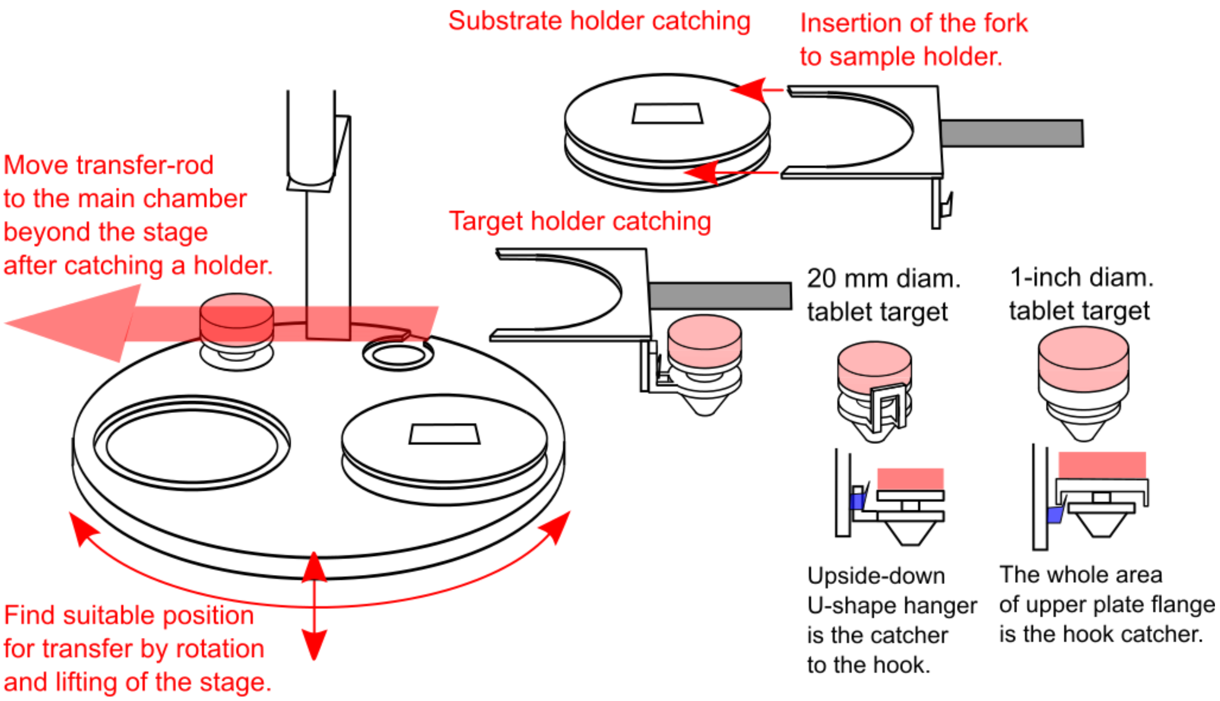

Figure 4:Catching substrate holder/target holder in load-lock chamber.

(2) Move the substrate stage/the target stage by handy console in the main chamber (see Figure 5) It is effective and easy way to use handy console for appropriate angle rotation in the substrate stage or the target stage when substrate holder/target holder transfer. The handy console operation is shown as follows.

a) Select a motorized control object by setting notch of the "Motor SEL" knob. And confirm that the indicator illuminates in accordance with the selected object.

b) Push the lower column lever down to left/right. Then confirm that the motorized motion of CW/CCW rotation, up/down motion or left/right motion occurs by following lever operation.

c) Push two levers down to the same direction at once for high-speed motion.

d) Utilize EMG (emergency), RST (reset), and ORG (origin: homing) operation buttons to solve trouble in motorized motion like gear stack and others.

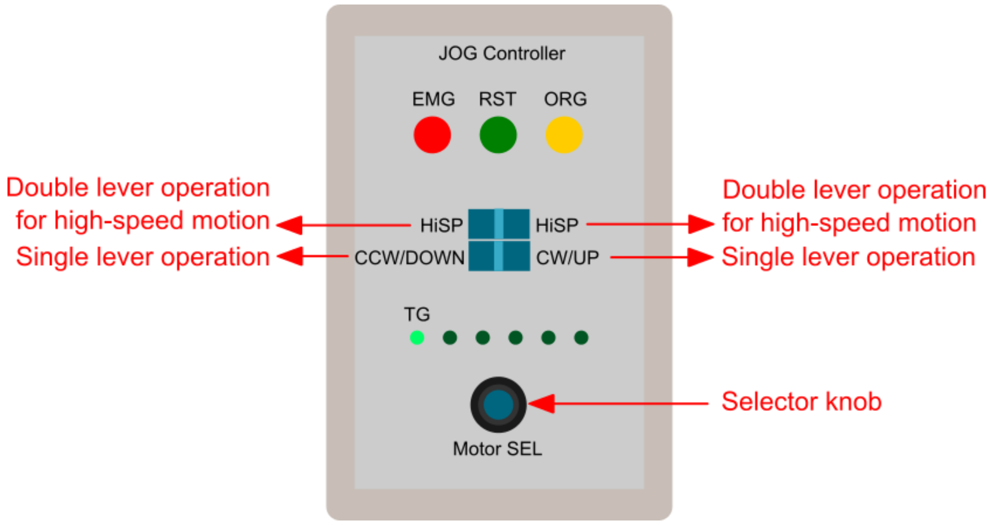

Figure 5:Substrate stage/target stage manipulation in main chamber by handy console.

(3) Transfer a substrate holder (see Figure 6)

a) Check that the L/L chamber is in UHV by vacuum pumping.

b) Move the substrate/target stocker with rotary motion and vertically motion by two handles beneath the L/L chamber to catch a substrate holder.

c) Check that the L/L chamber is in UHV by vacuum pumping.

d) Move the substrate/target stocker with rotary motion and vertically motion by two handles beneath the L/L chamber to catch a substrate holder.

e) Catch a substrate holder by insertion of folk to the substrate holder groove on side.

f) Open the isolation valve GV1.

g) Move the transfer-rod into the main chamber.

h) Move the substrate stage (laser shield) up relative to the substrate holder inside the main chamber for easy insertion of the holder to the stage.

i) Insert the substrate holder to the substrate stage.

j) Move the substrate stage slightly up to fix the round convex shape of the holder to the round hole of the stage.

k) Move off the transfer-rod.

l) Close the valve GV1.

m) Move the substrate stage down to the deposition position.

n) Do almost inverse procedure when picking up the substrate holder from the substrate stage in the main chamber.

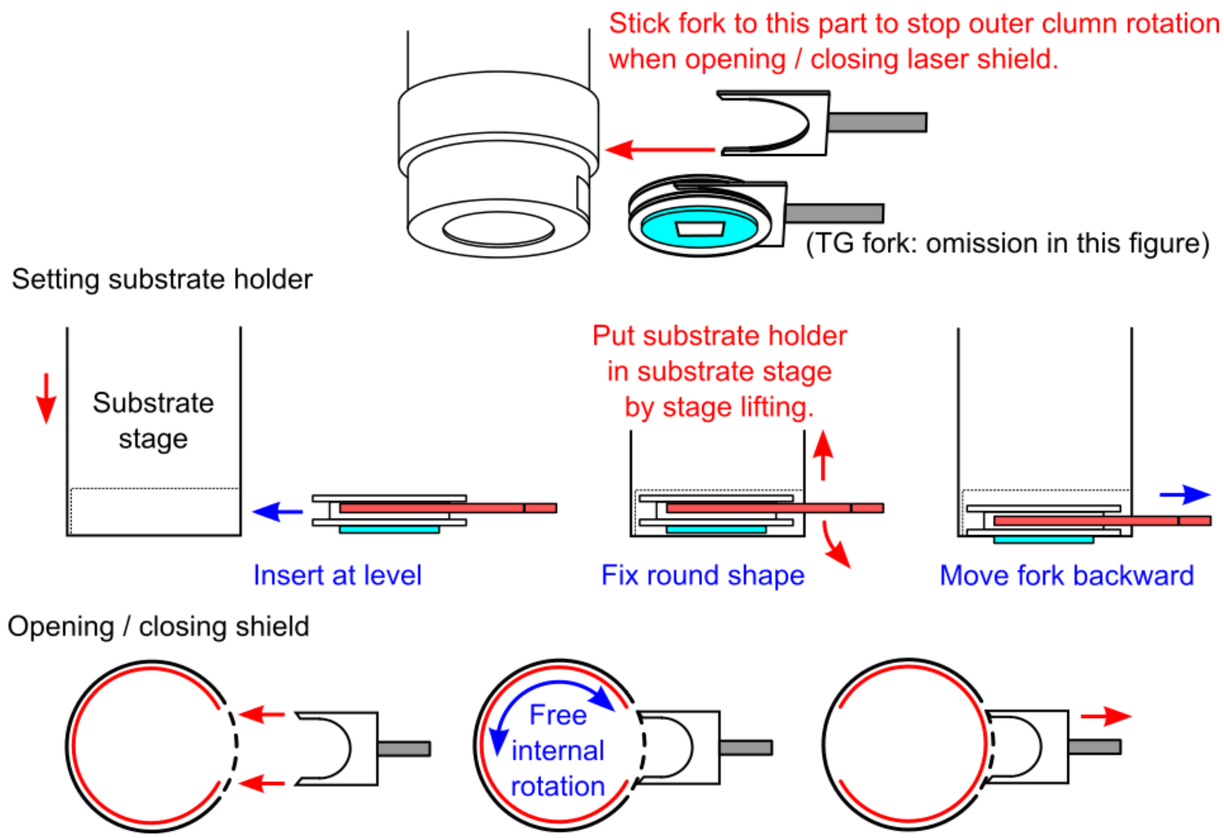

Figure 6:Substrate holder transfer to the substrate stage in the main chamber.

(4) Transfer a source target holder (see Figure 7)

a) Check that the L/L chamber is in UHV by vacuum pumping.

b) Move the substrate/target stocker with rotary motion and vertically motion by two handles beneath the L/L chamber to catch a substrate holder.

c) Catch a target holder by insertion of fork to the target holder flange.

d) Open the isolation valve GV1.

e) Move the transfer-rod into the main chamber.

f) Move the target stage up relative to the target holder inside the main chamber for easy insertion of the holder to the stage.

g) Check setting point from the viewpoint of cone-shape: convex in target holder and concave in target stage.

h) Move the target stage slightly up to fix at above setting position.

i) Move off the transfer-rod.

j) Close the valve GV1.

k) Move the target stage down to the deposition position.

l) Do almost inverse procedure when picking up the substrate holder from the substrate stage in the main chamber.

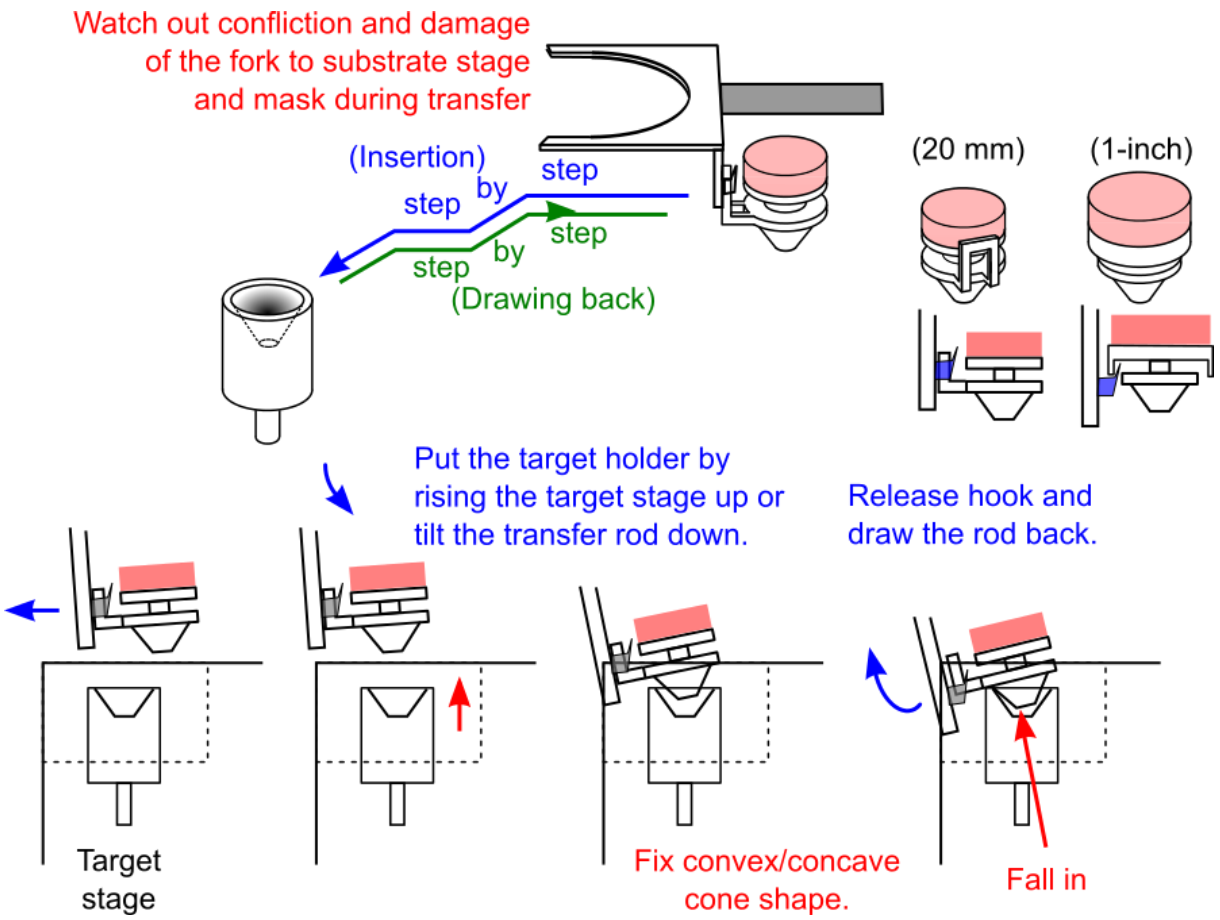

Figure 7:Target holder transfer to the target stage in the main chamber.

4.5 Process gas feeding

Process gas flow procedure is changed under desired condition as follows.

(High vacuum region up to 10^{-3} Pa)

(1) Feed process gas by opening the valve VLV.

(2) Make vacuum path through main pumping line of MV10.

(Low vacuum region up to 100 Pa)

(3) Make vacuum path through bypass pumping line of MV11. Close valve MV10.

(4) Feed process gas by opening the valve VLV or the valve PV.

(5) Set valve position of the manual valve near MV11 to adjust desired vacuum pressure.

(6) Measure the vacuum pressure by capacitance gauge (CDG).

(Large amount gas flow like post-annealing)

(7) Close the valves (GV) and MV2 for RHEED electron gun differential pumping line.

(8) Turn off the pump TMP2.

(9) Close the valve FV2 after stopping above pumps.

(10) Close valve CV1 near capacitance gauge (CV1).

(11) Close the valve MV10 in main pumping line and close the valve MV11 in bypass pumping line.

Main chamber is not pumped. There is a required condition in which the TMP1 is running and one valve MV10 or MV11 is open for post-annealing process with laser-diode substrate heating.

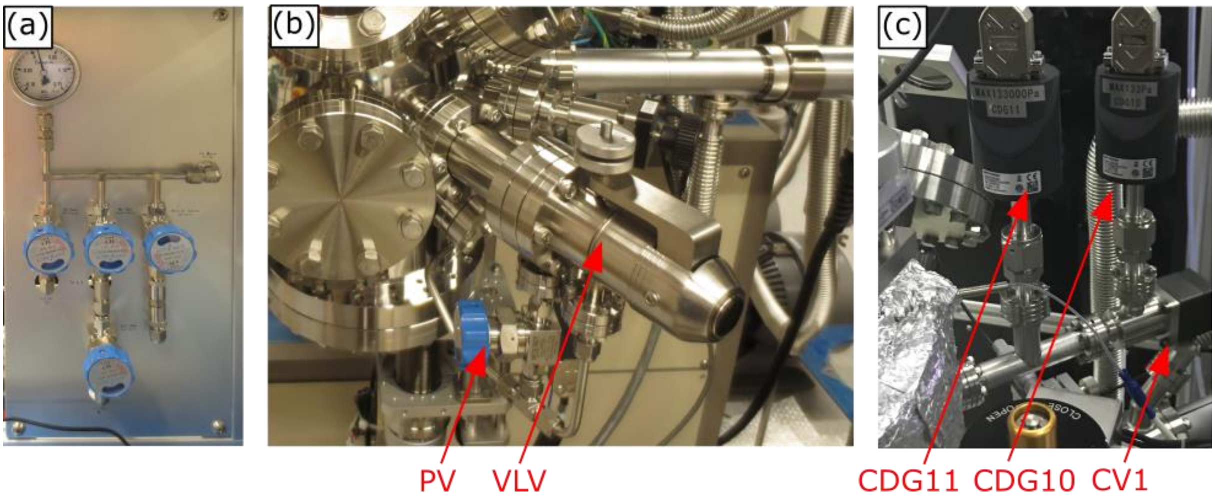

Figure 8:Gas feeding part in the PLD system (main chamber and panel).

(12) Feed process gas by opening the valve PV1.

(13) Measure the vacuum pressure in the main chamber by CDG.

(14) Pay attention for positive pressure. Main chamber does not have escape path to decrease positive

pressure.

(15) When finishing the process, restart vacuum pumping.

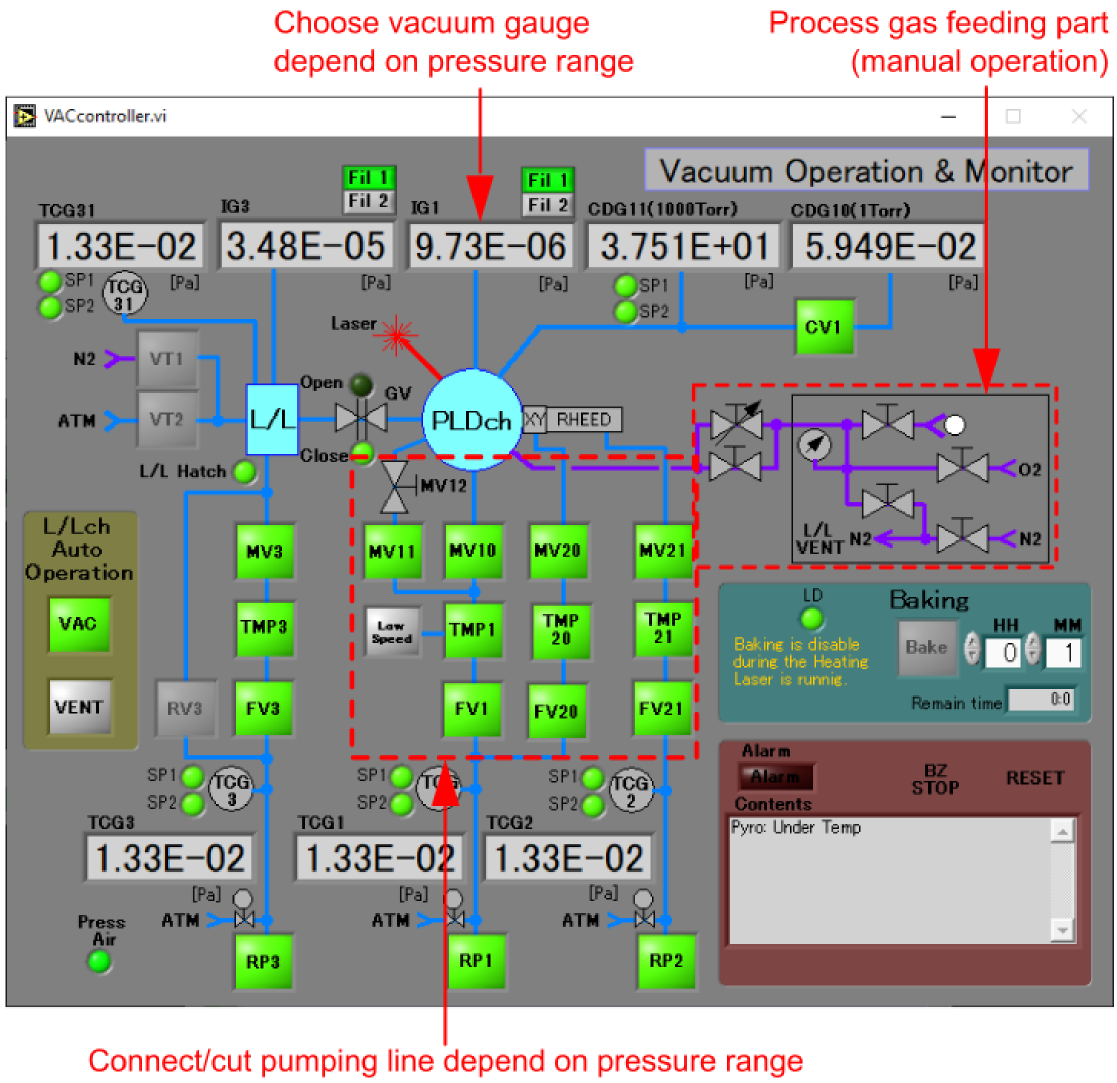

Figure 9:Gas feeding part in control software.

4.6 Substrate heating by infrared (IR) laser-diode

(Initializing)

(1) Turn on the cooling water circulator. First turn on the main switch on side panel and check that the front display shows a temperature after a few seconds. Then push “RUN” button of front panel for a few seconds. And check water-flow rate and the water purity.

(2) Turn on main key switch in laser-diode power supply 5 minutes after above.

(3) Check that power indicator illuminates and “LD on” indicator emits green light. Then check that all alarm/interlock-indicators do not illuminate.

(4) Select “EXT” mode of laser-diode control switch in the case of PC control.

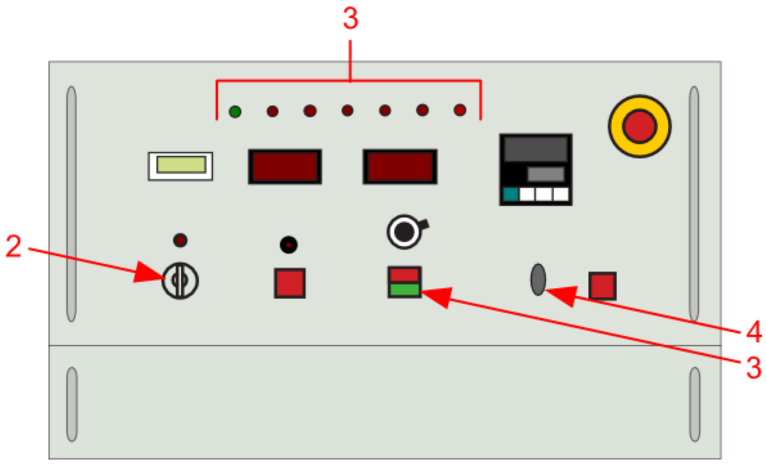

Figure 10:Substrate heating operation on front panel of laser-diode power supply.

(Laser-diode control in driving current mode on PC control software)

(5) Check that all alarm indicators do not light.

(6) Turn to “Active” in the LD active toggle switch of the control software. Then check that the “Ready” indicator shows green.

(7) Turn on “LD on” toggle switch of the control software. Then check that the “LD on” indicator shows green.

(8) Turn on “Threshold” toggle switch of the control software to start laser oscillation. Then check that the laser-diode driving current is set to threshold current according to initial setting for example at around 8-9 A. Check that the “Stand-by” indicator shows green.

(9) Set desired current value by operating LD current adjuster with slider, cursor or direct input of the value in the control software.

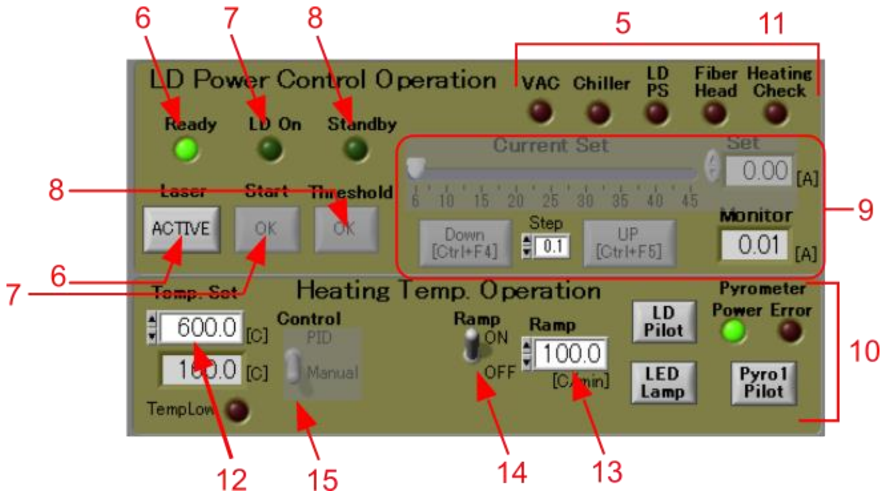

Figure 11:Substrate heating operation on PC control software.

(Laser-diode control by temperature PID mode in PLD control software)

(10) Check that two pyrometers work well and show proper temperature value as contrasted with the laser-diode driving current without error.

Control object of temperatrue PID is the value of pyrometer #1.

(11) Check that "Heating Check" alarm indicator does not light.

(12) Set desired substrate temperature.

(13) Set ramping rate to the goal temperature in temperature control.

(14) Turn to "ON" of Ramp mode switch.

(15) Turn to "PID" in the temperature control toggle switch.

(Shutdown)

(16) Turn off "Threshold" toggle switch of the control software to start laser oscillation. Then check the laser-diode driving-current decreases from the threshold current to zero gradually.

(17) Turn off "LD on" toggle switch of the control software.

(18) Turn off LD active toggle switch of the control software.

(19) Turn off the main key switch.

(20) Turn off the chiller 30 minutes after turning off the main key switch.

4.7 Deposition by individual operation

In the PLD system deposition sequence, choosing target or ablation laser trigger is controlled with the software. In the software each deposition factor can be controlled individually. Figure 33 shows other control items of choosing target, substrate rotation or laser oscillation.

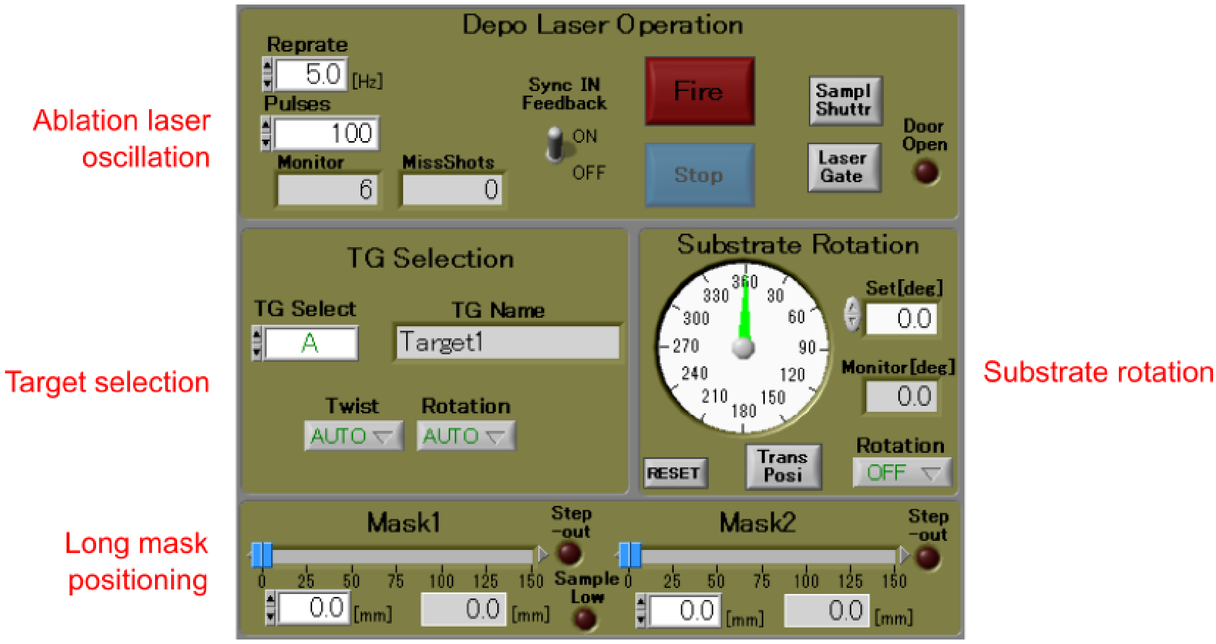

Figure 12:Other control objects on PC control software.

4.8 Shutdown PLD system

PC software operation is similar to that in start-up procedure described in Sec 4.1. Refer the sentences and the figure.

(Main chamber with RHEED differential pumping line)

(1) Open the gate valve GV1 when user wants to do system maintenance with whole vent.

(2) Turn off a filament of IG1 vacuum gauge. Click green "Fil 1/2" button of IG1: off status of the IG1 is gray in the button indicator.

(3) Close the valve CV1 near capacitance manometer CDG10: close status of the CV1 is gray in the button indicator.

(4) Close the valves MV10: close status of the MV10 is gray in the button indicator.

(5) Close the valves MV11: close status of the MV11 is gray in the button indicator.

(6) Close the valve MV12 manually.

(7) Close the valves MV20 and MV21 for RHEED e-gun differential pumping line: close status of the valves is gray in the button indicator.

(8) Turn off the pump TMP1 and pumps TMP20 and TMP21. And check the display in the TMPs' controller.

(9) Close the valve FV1 and valves FV20 and FV21: close status of the valves is gray in the button indicator.

(10) Turn off the pump RP2: off status of the RP2 is gray in the button indicator.

(11) Turn off the pump RP1: off status of the RP1 is gray in the button indicator.

(Load-lock chamber)

(12) Turn off a filament of IG3 vacuum gauge. Click green "Fil 1/2" button of IG3: off status of the IG3 is gray in the button indicator.

(13) Close valve MV3: close status of the MV3 is gray in the button indicator.

(14) Turn off the pump TMP3. And check the display in the TMP3 controller.

(15) Close the valve FV3: close status of the FV3 is gray in the button indicator.

(16) Turn off the pump RP3: off status of the RP3 is gray in the button indicator.

5 Automatic control of deposition process by script

5.1 Basic concept of script

Script is automatic control function for deposition sequence. The script includes some operations like firing ablation laser, target manipulation, substrate temperature control, or others. And the script also includes some parameter settings. These operations and parameter settings are formed to **“commands”** in the script. Moreover the script has **construction sentences** like “loop” to realize repetition of deposition process. The commands and the construction sentences are effectively used for the automation.

A script file is saved in PC as text format file. So the script file is easy to modify by this software or other editor software.

5.2 Script editing and executing environment

A sub-window for script editing/running has following window (screen) layout.

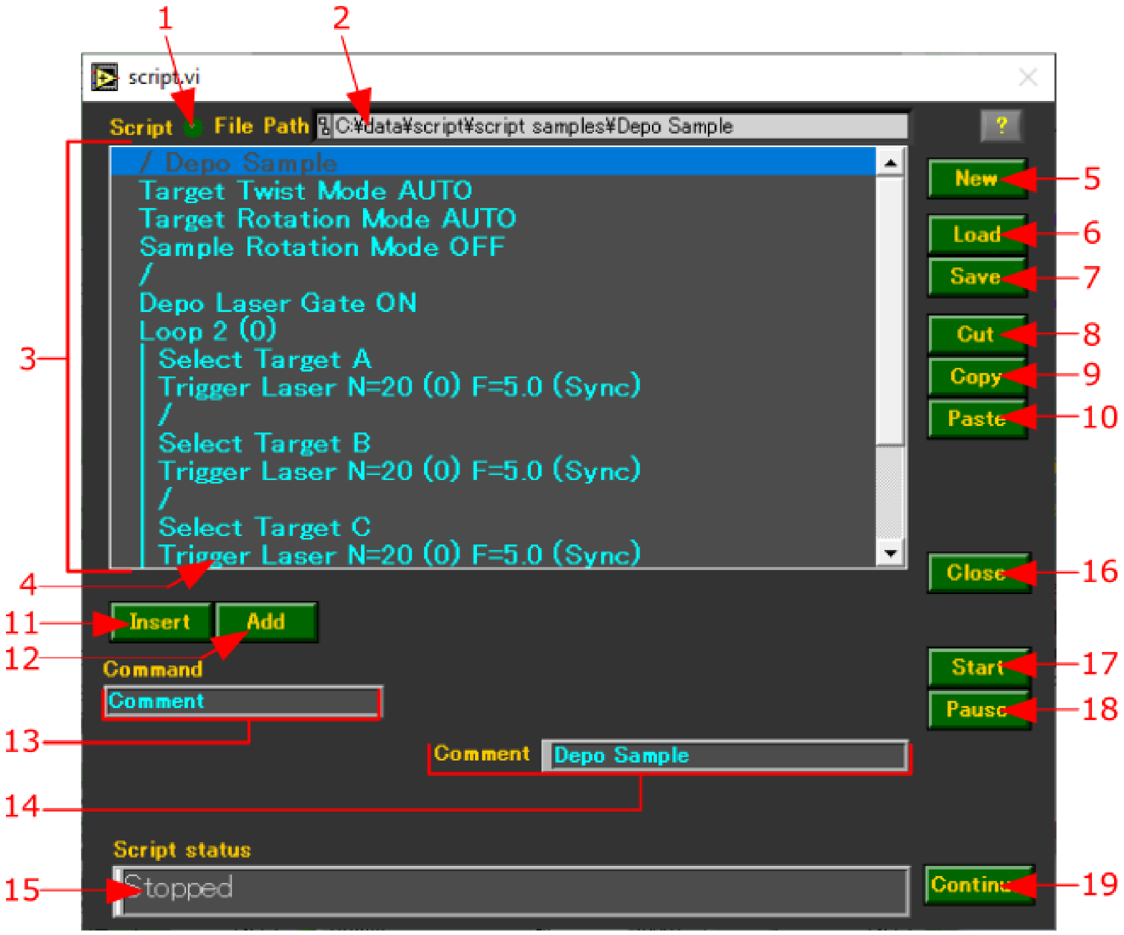

Figure 13:Script editing/executing sub-window in PC control software.

Table 1:Function in script editing/executing sub-window.

1	Indicator	The indicator lights when the script file is modified in the command edit box (#3) to prompt for file save.
2	Script file path set box	The box shows a path to access the script file.
3	Command edit box	The box is a kind of editor for script.
4	Current command cursor	The cursor is shown in blue line to highlight. When script-editing the cursor points out the command line to modify. When script-running the cursor points out the command line to operate currently.
5	New (button)	The editing-command creates new script in command edit box. A message to save former script appears just before the operation.
6	Load (button)	Load script file. The button opens a sub-window to load a script file to command edit box.
7	Save (button)	Save script file. Script in the command edit box is saved to a PC file. When the script is created by “New” editing-command the script is saved to a new file with setting new file name. When the script is modified from loaded file by “Load” editing-command the script is saved by overwriting to the same file namewith confirmation message.
8	Cut (button)	The editing-command cut the command line(s) at current command cursor and store to clip-board.
9	Copy (button)	The editing-command copy the command line(s) at current command cursor and store to clip-board.
10	Paste (button)	The editing-command pastes the command line(s) at current command cursor from clip-board.
11	Insert (button)	The editing-command creates a new command line above current command cursor. Initial setting of the command line is “NOP” script command and the command line can be modified with pull-down list.
12	Add (button)	The editing-command creates a new command line below current command cursor. Initial setting of the command line is “NOP” script command and the command line can be modified with pull-down command list.
13	Command (body)	The box shows the script command at current command cursor when script-editing. And the script command is modified with pull-down command list.
14	Command parameter & Postfix	The box shows the script command parameters and postfix-options at current command cursor. The parameters and options are modified in the box when script-editing. And parameters are shown as current status when script-running.
15	Script status	The box shows running status of the script.
16	Close (button)	The button closes the sub-window.
17	Start (button)	The command executes the script from first command line.
18	Pause (button)	The command interrupts executing the script.
19	Continue (button)	The command restarts the script at the command line where interruption by “Pause” command.

5.3 Modifying script by text-base editor software

The saved script file has text-based format and has the file name without extension. So the file can be reviewed and modified by editor software.

6 Command reference

6.1 Basic sentence pattern of script

Script is constructed by “commands” which control each unit in PLD system (Laser MBE system) and “construction sentences” which define repetition of process or other sequence controls. All commands and construction sentences are described by following format.

Command/ Construction sentence	Parameter1	Parameter2	…	Wait/Sync Postfix options

(1) **Command / construction sentences** : the phrase is expressed by simple and easy understanding keywords to control each unit in PLD (Laser MBE) system.

(2) **Parameters** : these are fields to define proper values for unit control. Zero to four parameter fields are needed in accordance to commands and their functions.

(3) **Sync/Wait postfix options** : these options are added when the command requires synchronized operation or waiting until finishing the operation.

6.2 Commands for source target manipulation

Command (Syntax/Example)	Wait	Sync	Function
Seek Targets Home **Command syntax** Seek Targets Home **Example** Seek Targets Home	Enable	Disable	This command does homing operation of target manipulator. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation. (see “Homing” operation and use of handy console)
Select Target **Command syntax** Select Target [*Alphabet notation of a target*] **Example** Select Target B	Enable	Disable	This command moves the target, which you choose, to deposition position. ***Alphabet notation*** (A, B, …) + **"Clear"** is needed when you choose the target. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.
Target Rotation Mode **Command syntax** Target Rotation Mode [*OFF* / *ON* / *Auto*] **Example** Target Rotation Mode ON	Disable	Disable	This command controls target spinning with following modes. ***OFF*** : this setting quits target spinning. ***ON***: this setting begins target spinning. ***AUTO***: this setting begins or quits target spinning automatically. Target spinning is carried out during executing “Trigger laser” command.
Target Twist Mode **Command syntax** Target Twist Mode [*OFF* / *ON* / *Auto*] **Example** Target Twist Mode ON	Disable	Disable	This command controls target twist with following modes. ***OFF*** : this setting quits target twist. ***ON***: this setting begins target twist. ***AUTO***: this setting begins or quits target twist automatically. Target twist is carried out during executing “Trigger laser” command.

(Continued)

Command (Syntax/Example)	Wait	Sync	Function
Seek TG-Z Home **Command syntax** Seek TG-Z Home **Example** Seek TG-Z Home	Enable	Disable	This command does homing operation of Z- (longitudinal) position in target stage. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation. (see "Homing" operation and use of handy console)
Set TG-Z Position **Command syntax** Set TG-Z Position [*POS*] **Example** Set TG-Z Position 25mm	Enable	Disable	This command moves the target stage to set Z- (longitudinal position) value. ***POS***: set Z-value. It is absolute value from 0 to 40 mm. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.
Move TG-Z **Command syntax** Move TG-Z [*STR*] **Example** Move TG-Z 5mm	Enable	Disable	This command moves the target stage to set Z- (longitudinal position) value. ***STR***: set Z-value. It is relative value to the present position. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.

6.3 Commands for UV ablation laser oscillation

Command (Syntax/Example)	Wait	Sync	Function
Trigger Laser **Command syntax** Trigger Laser [*set pulse number*] [*set pulse rep. rate*] **Example** Trigger Laser 1000 10	Enable	Enable	This command emits ablation laser pulse with ***set pulse number*** and ***set pulse rep. rate***. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation, emission (oscillation) of ablation laser pulse. When **Sync** postfix option is set, this command takes sync out signal from ablation laser.
Depo Laser Gate **Command syntax** Depo Laser Gate [*OFF* / *ON*] **Example** Depo Laser Gate ON	Enable	Disable	This command opens / closes gate shutter. ***OFF***: closes the gate shutter. ***ON***: opens the gate shutter. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.
Sample Shutter **Command syntax** Sample Shutter [*OFF* / *ON*] **Example** Sample Shutter ON	Enable	Disable	This command opens / closes sample shutter. ***OFF***: closes the sample shutter. ***ON***: opens the sample shutter. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.

(Continued)

Command (Syntax/Example)	Wait	Sync	Function
Combi Laser **Command syntax** Combi [*set pulse number*] [*set pulse repetition rate*] [*Mask 1 stroke*] [*Mask 2 stroke*] **Example** Combi N=10(0) F=10.0 M1=0.00 M2=0.00 *See section 6-5 in particular "Move mask" command.*	Enable	Enable	This command synchronously control of ablation laser emission with ***set pulse number / set repetition rate*** and M1- (and/or M2-) mask motion with ***stroke***. ***Stroke*** value is relative including minus. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation, emission (oscillation) of ablation laser pulse. When **Sync** postfix option is set, this command takes sync out signal from ablation laser.

6.4 Commands for ablation laser beam focusing

Command (Syntax/Example)	Wait	Sync	Function
Seek Lens Home **Command syntax** Seek Lens Home **Example** Seek Lens Home	Enable	Disable	This command does homing operation of lens position in laser optics box. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.
Set Lens Position **Command syntax** Set Lens Position [*Pos*] **Example** Set Lens Position 25mm	Enable	Disable	This command moves the focusing lens position to set position value. ***POS***: set lens position. It is absolute value from 0 to 100 mm. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.
Move Lens **Command syntax** Move Lens [*STR*] **Example** Move Lens 20mm	Enable	Enable	This command moves the focusing lens position to set stroke value. ***STR***: lens position stroke. It is relative value involving negative from the last position. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.

6.5 Commands for attenuator control

Command (Syntax/Example)	Wait	Sync	Function
Seek ATT Home **Command syntax** Seek ATT Home **Example** Seek ATT Home	Enable	Disable	This command does homing operation of lens position in laser optics box. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.
Set ATT Angle **Command syntax** Set ATT Angle [*Angle*] **Example** Set ATT Angle 30deg	Enable	Disable	This command rotates the attenuator optics to set angle value. ***Angle***: set rotation angle. It is absolute value from 0 to 45 deg. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.
Rotate ATT **Command syntax** Rotate ATT [*Angle*] **Example** Rotate ATT -10deg	Enable	Enable	TThis command rotates the attenuator optics within set angle value. ***Angle***: rotation angle. It is relative value involving negative from the last angle position. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.

6.6 Commands for sample rotation

Command (Syntax/Example)	Wait	Sync	Function
Set Sample Position **Command syntax** Set Sample Position [*set rotation angle*] **Example** Set Sample Position 45.0	Enable	Enable	This command rotates the sample (substrate) to set rotation angle. The ***set rotation angle*** is absolute value from 0 to 359.9 deg. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the sample rotation. When **Sync** postfix option is set, this command acts synchronously with the sync-out signal.
Rotate Sample **Command syntax** Rotate Sample [*set rotation angle*] **Example** Rotate Sample 60	Enable	Enable	This command rotates the sample (substrate) with set rotation angle. The ***set rotation angle*** is relative value to present angle position. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the sample rotation. When **Sync** postfix option is set, this command acts synchronously with the sync-out signal.
Sample Rotation Mode **Command syntax** Sample Rotation Mode [*OFF* / *ON* / *Auto*] **Example** Sample Rotation Mode AUTO	Disable	Disable	This command operates endless sample (substrate) rotation with following spin setting modes. ***OFF*** : this setting quits endless sample rotation. ***ON***: this setting begins endless sample rotation. ***AUTO***: this setting begins or quits endless sample rotation automatically. Endless sample rotation begins just before executing “trigger laser” command and quit after finishing ablation laser oscillation.

6.7 Commands for combinatorial mask positioning

Command (Syntax/Example)	Wait	Sync	Function
Mask Speed **Command syntax** Mask Speed [*M1* / *M2*] [*Low*] [*High*] [*Acc*] **Example** Mask Speed M1 Low=1 High=100 Accel=500	Disable	Disable	This command defines speed parameters in combinatorial mask positioning unit. ***M1 / M2*** : the parameter chooses a mask M1 : Mask 1, M2 : Mask 2 ***Low*** : (Reserved parameter) it sets starting speed. ***High*** : the parameter sets maximum speed. An example data of 12000pps equals to 1.0cm/s in Mask 1 case. ***Acc*** : (Reserved parameter) it sets acceleration speed.
Seek Mask Home **Command syntax** Seek Mask Home [*M1* / *M2*] **Example** Seek Mask Home M1	Enable	Disable	This command does homing operation of mask position. ***M1 / M2*** : the parameter chooses a mask M1 : Mask 1, M2 : Mask 2 When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation. (see "Homing" operation and use of handy console)
Set Mask Position **Command syntax** Set Mask Position [*M1* / *M2*] [*POS*] **Example** Set Mask Position M1=0.00	Enable	Disable	This command moves the combinatorial mask to set position value. ***M1 / M2*** : the parameter chooses a mask M1 : Mask 1, M2 : Mask 2 ***POS***: set position value. It is absolute value from 0 to 150 mm. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation.

(Continued)

Command (Syntax/Example)	Wait	Sync	Function
<u>Move Mask</u> **Command syntax** Move Mask [*M1* / *M2*] [*STR*] **Example** Move Mask M1=0.00	Enable	Enable	This command moves the combinatorial mask to set position value. ***M1 / M2*** : the parameter chooses a mask M1 : Mask 1, M2 : Mask 2 ***STR***: set position value. It is relative value to the present position. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation. When **Sync** postfix option is set, this command acts synchronously with the sync-out signal.
<u>Combi Laser</u> **Command syntax** Combi [*set pulse number*] [*set pulse repetition rate*] [*Mask 1 stroke*] [*Mask 2 stroke*] **Example** Combi N=10(0) F=10.0 M1=0.00 M2=0.00 *See section 6-3 in particular "Trigger laser" command.*	Enable	Enable	This command synchronously control of ablation laser emission with ***set pulse number / set repetition rate*** and M1- (and/or M2-) mask motion with ***stroke***. ***Stroke*** value is relative including minus. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation, emission (oscillation) of ablation laser pulse. When **Sync** postfix option is set, this command takes sync out signal from ablation laser.

6.8 Commands for laser-diode substrate heating

Command (Syntax/Example)	Wait	Sync	Function
Heating Laser Lock **Command syntax** Heating Laser Lock [*Locked* / *Active*] **Example** Heating Laser Lock Active	Enable	Disable	This command allows or forbids laser-diode driving current control operations and PID temperature control operations. ***Locked***: this setting forbids the operations. ***Active***: this setting allows the operations. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the Locked/Active operation.
Heating Laser Pointer **Command syntax** Heating Laser Pointer [*OFF* / *ON*] **Example** Heating Laser Pointer ON	Disable	Disable	This command turns red pilot-laser light, which is used for collimation of infrared laser light in substrate heating, on or off. ***OFF***: this setting turns pilot-laser light off. ***ON***: this setting turns pilot-laser light on.
Heating Laser **Command syntax** Heating Laser [*OFF* / *ON*] **Example** Heating Laser ON	Enable	Disable	This command turns LD switch on or off. ***OFF***: this setting turns LD switch off. ***ON***: this setting turns LD switch on. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the LD on/off operation.
Heating Laser Threshold **Command syntax** Heating Laser Threshold [*OFF* / *ON*] **Example** Heating Laser Threshold ON	Disable	Disable	This command warms LD module up at starting current for oscillation. ***OFF***: this setting quit LD module warming up. ***ON***: this setting begins LD module warming up.

(Continued)

Command (Syntax/Example)	Wait	Sync	Function
Set Heating Current **Command syntax** Set Heating Current [*set current value*] **Example** Set Heating Current 14.0A	Enable	Disable	This command adjusts laser-diode driving current to ***set current value***. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the LD current set.
Set Minimum Current **Command syntax** Set Minimum Current [*set current value*] **Example** Set Current Limit 10.0A	Disable	Disable	This command defines lower limit of laser-diode driving-current to ***set current value***. The *set current value* should be than the threshold current for laser oscillation. *Corresponding numerical setting is in "PID" setting sub-window.*
Set Maximum Current **Command syntax** Set Maximum Current [*set current vale*] **Example** Set Starting Current 43.0A	Disable	Disable	This command defines upper limit of laser-diode driving current to ***set current value***. The *set current value* should be less than the output limit of the power supply. *Corresponding numerical setting is in "PID" setting sub-window.*
Temperature Set **Command syntax** Temperature Set [*set temperature value*] **Example** Temperature Set 800degC	Enable	Disable	This command sets goal temperature (***set temperature value***) when the substrate heating is in PID temperature control mode. When **Wait** postfix option is set, this command does not handle the control to a next command until finishing the operation, i.e., substrate temperature reaches the goal temperature

(Continued)

Command (Syntax/Example)	Wait	Sync	Function
Temperature Tolerance **Command syntax** Temperature Tolerance [*set tolerance value*] **Example** Temperature Tolerance 10degC	Disable	Disable	This command defines temperature tolerance to ***set tolerance value*** at goal temperature when the substrate heating is in PID temperature control mode. Set value is denoted in unit of deg C.
Temperature Ramp **Command syntax** Temperature Ramp [*ON* / *OFF*] [*set ramp-rate value*] **Example** Temperature Ramp ON 100	Disable	Disable	*This command sets ramp-rate when the substrate heating is in PID* temperature control mode. The ***set ramp-rate value*** is set in unit of deg C/ min. ***OFF***: this setting does not control substrate temperature. ***ON***: this setting controls substrate temperature to goal temperature with ***set ramp-rate value***.
Temperature Control **Command syntax** Temperature Control [*OFF* / *ON*] **Example** Temperature Control ON	Disable	Disable	*This command changes temperature control mode in substrate heating.* ***OFF***: this setting sets manual mode. The PLD system only increases/decreases laser-diode driving current for substrate heating. ***ON***: this setting sets PID temperature control mode. The PLD system automatically raises up or lowers substrate temperature by PID feedback control with pyrometer temperature measurement.
LED Lamp **Command syntax** LED Lamp [*ON* / *OFF*] **Example** LED Lamp ON	Disable	Disable	This command turns white LED light on or off to see bright image of substrate-backside. ***OFF***: this setting turns LED light off. ***ON***: this setting turns LED light on.

6.9 Data logging commands and other commands

Command (Syntax/Example)	Wait	Sync	Function
Set Log Interval **Command syntax** Set Log Interval [*ITSEC*] **Example** Set Log Interval 10	Disable	Disable	This command set a time interval of data acquisition in chart-type monitor, namely LD current monitor, substrate temperature monitor, and pressure monitor at right side of the main window. The command requires a following parameter field. ***ITSEC***: this parameter field sets the time interval of data acquisition in second.
Data Logging **Command syntax** Data Logging [*OFF* / *ON*] [*LOGFILE*] **Example** Data Logging ON March4.txt	Disable	Disable	This command writes monitoring data of LD current, substrate temperature, and main chamber pressure out to a log file. The command requires two following parameter fields. ***OFF***: this setting quits writing data. ***ON***: this setting begins writing data. ***LOGFILE***: this parameter field sets file name of log-file.
Beep **Command syntax** Beep **Example** Beep	Disable	Disable	*This command beeps for a moment.*
Video Select **Command syntax** Video Select [CCD1 / CCD2 / CCD3] **Example** Video Select CCD1	Disable	Disable	This command choose video a image to show a window, part of PC screen from 3 CCDs.

6.10 Construction sentences

Command (Syntax/Example)	Wait	Sync	Function
<u>NOP</u> **Command syntax** NOP **Example** NOP	Disable	Disable	This command only progresses to next command. Notation of the command means "no operation". *This command appears first when editing a new command line with "**Add**" or "**Insert**" operation in script editing/executing sub-window.*
<u>Comment</u> **Command syntax** / *Remark sentences* **Example** / You can write comments to the end of the line.	Disable	Disable	This command shows comment to clarify function of the script. Normal expression of the *Comment command* is "/" at the beginning of the line. User can write comments in the line to the end.
<u>Wait</u> **Command syntax** Wait [*Set time*] **Example** Wait 10sec	Disable	Disable	*This command waits for set time. The command requires a following* parameter field. ***Set time***: this parameter field sets waiting time in second.
<u>Wait for Continue</u> **Command syntax** Wait for Continue **Example** Wait for Continue	Disable	Disable	This command interrupts executing a script until **"Continue" button** in script editing/executing sub-window is pressed.

(Continued)

Command (Syntax/Example)	Wait	Sync	Function
<u>Loop / Loop End</u> **Command syntax** Loop [*Repetition times*] | (Commands recurrently executing) Loop End **Example** Loop 50(0) | | | Loop End	Disable	Disable	These construction sentences, “Loop” and “Loop End”, repeats commands in the part from “Loop” to “Loop End” set repetition times. The Loop construction sentence requires a following parameter field. ***Repetition times***: this parameter field sets repetition times.

7 Sample script

7.1 Example #1: Fabrication of multilayered structure by 2 kinds of thin film materials with substrate heating

This is an example of script to make multilayered structure like superlattice by 2 kinds of thin film materials. In this case iteration is five times. Temperature control is also achieved during the deposition process. Deposition diagram with the condition is shown in Figure 18.

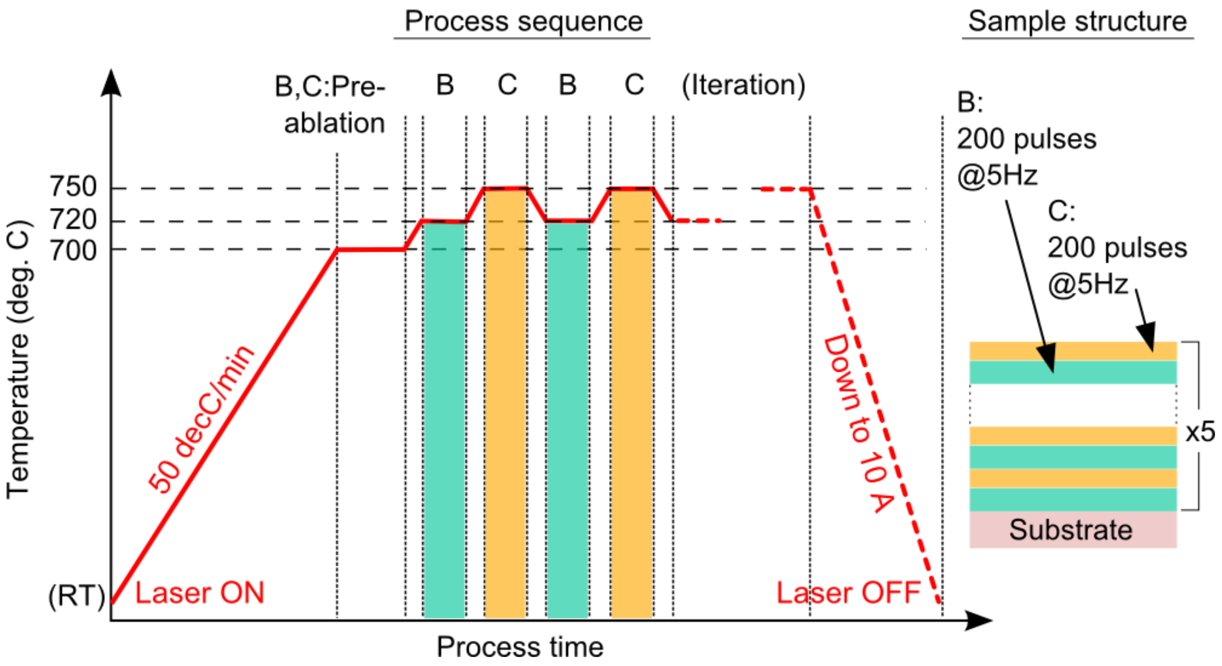

Figure 14:Process sequence and sample structure of script example #1.

<u>Script example</u>

/ Start up diode-laser for substrate heating.

Heating Laser Lock ACTIVE

Heating Laser ON

Heating Laser Threshold ON

Wait 10 sec (0)

/ Increase in laser-diode current step by step

Set Heating Current 10.00

Wait 20 sec (0)

Set Heating Current 11.00

Wait 20 sec (0)

/ PID temperature control to 700 degC

Temperature Set 180.0 (Nowait)

Temperature Control PID

Temperature Tolerance 2.0

Temperature Ramp 50.0

Temperature Set 700.0

Wait for Continue

/ Pre-ablation

Sample Shutter OFF

Select Target B

Trigger Laser N=3000 (0) F=10 (Sync)

Select Target C

Trigger Laser N=3000 (0) F=10 (Sync)

/ Iteration of B or C film deposition at each temperature.

Loop 5 (0)

| Temperature Set 720.0

| Select Target B

| Sample Shutter ON

| Trigger Laser N=200 (0) F=5 (Sync)

| Sample Shutter OFF

| Temperature Set 750.0

| Select Target C

| Sample Shutter ON

| Trigger Laser N=200 (0) F=5 (Sync)

| Sample Shutter OFF

Loop End

/ Cool the substrate temperature down.

Temperature Control Manual

Set Heating Current 10.00

Wait 10 sec (0)

/ Shut the laser-diode down

Heating Laser Threshold OFF

Heating Laser OFF

Heating Laser Lock LOCKED

7.2 Example #2: Fabrication of 3-elements composition spread film by using trapezoid moving mask/triangle contact mask

Triangle-shaped composition spread film can be made by using trapezoidal moving mask and triangle contact mask. In this case dynamic shading during deposition, synchronous mask motion to ablation laser oscillation, is effectively used to make gradation. The sample structure and an example of mask position /stroke are shown in Figure 19.

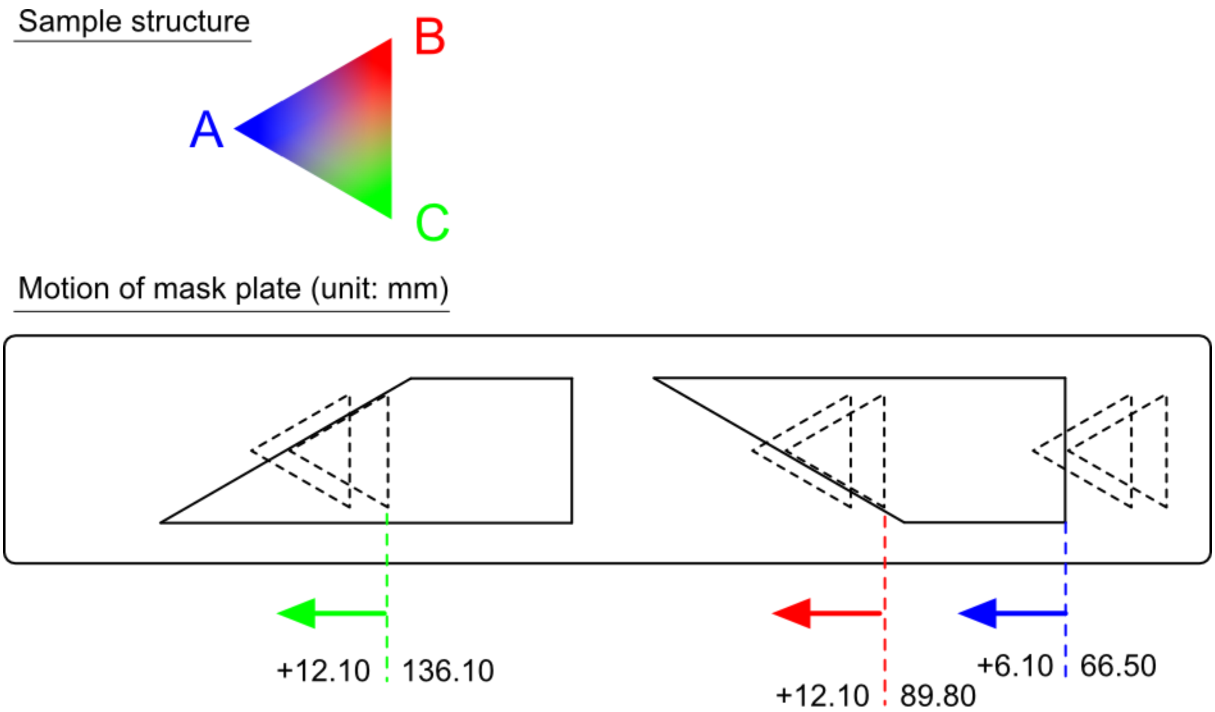

Figure 15:Sample mapping and mask position/stroke of script example #2.

Script example

/ Initial and high-speed setting of mask speed

Mask Speed M2 Low=1 High=25000 Accel=4000

Set Mask Position M2=60.00

Wait for Continue

Target Rotation Mode AUTO

Target Twist Mode AUTO

/ Pre-ablation

Select Target A

Trigger Laser N=3000 (100) F=10 (Sync)

Select Target B

Trigger Laser N=3000 (100) F=10 (Sync)

Select Target C

Trigger Laser N=3000 (100) F=10 (Sync)

Wait for Continue

/ Iteration of deposition with mask shading for fabrication of 3 elements composition mapping.

/ using new command Combi Laser

Loop 3 (0)

| Mask Speed M2 Low=1 High=25000 Accel=4000

| Set Mask Position M2=66.50

| Select Target A

| Combi N=100 (0) F=10 M1=0.0 M2=6.10 (Nowait)

| Mask Speed M2 Low=1 High=25000 Accel=4000

| Set Mask Position M2=89.80

| Select Target B

| Combi N=100 (0) F=10 M1=0.0 M2=12.10 (Nowait)

| Mask Speed M2 Low=1 High=25000 Accel=4000

| Set Mask Position M2=136.10

| Select Target C

| Combi N=100 (0) F=5 M1=0.0 M2=12.10 (Nowait)

Loop End

Mask Speed M2 Low=1 High=25000 Accel=4000

Set Mask Position M2=0.00

Operation manual of PAC-LMBE system

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